

Indices: The Laws of Powers

WorksheetHub AI
Mathematics · Grade 7

Unit 1 · Number

Medium

Indices

Name: _____

Date: _____

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Remember

An **index** (or power) tells you how many times to multiply a number by itself. In 2^5 , the base is 2 and the index is 5: $2 \times 2 \times 2 \times 2 \times 2 = 32$.

The index laws

Multiply

$$a^m \times a^n = a^{m+n}$$
$$2^3 \times 2^2 = 2^5$$

Divide

$$a^m \div a^n = a^{m-n}$$
$$5^6 \div 5^2 = 5^4$$

Power of a power

$$(a^m)^n = a^{m \times n}$$
$$(3^2)^3 = 3^6$$

Power of zero

$$a^0 = 1$$
$$9^0 = 1$$

Negative power

$$a^{-n} = \frac{1}{a^n}$$
$$2^{-3} = \frac{1}{8}$$

Power of one

$$a^1 = a$$
$$7^1 = 7$$



Watch out

The laws for multiplying and dividing only work when the **bases are the same**.

A. Multiply and divide (write as a single power)

1 $3^5 \times 3^2$

Answer:

2 $5^8 \div 5^3$

Answer:

3 $x^4 \times x^6$

Answer:

4 $y^9 \div y^4$

Answer:

5 $7^3 \times 7^1$

Answer:

6 $a^2 \times a^3 \times a^4$

Answer:

B. Power of a power

7 $(2^3)^4$

Answer:

8 $(m^2)^5$

Answer:

9 $(10^2)^3$
Answer:

C. Zero and negative powers

10 $9^0 = ?$
Answer:

11 $4^0 + 3^0 = ?$
Answer:

12 $2^{-3} = ?$ (write as a fraction)
Answer:

13 $10^{-2} = ?$ (write as a fraction)
Answer:

D. Calculate

14 $2^3 \times 2^2 = ?$ (give the number)
Answer:

15 $10^6 \div 10^4 = ?$ (give the number)
Answer:

16 $(3^2)^2 = ?$
Answer:

★ Challenge

17 Find x: $2^x = 64$
Answer:

18 Simplify $\frac{3^4 \times 3^2}{3^3}$
Answer:

19 Which is bigger: 2^{10} or 10^2 ? Show your working.
Answer: